

Technical Report: Domain Name Generation with LLM Fine-Tuning & Evaluation

1. Objective

The goal of this project is to develop and evaluate multiple LLM-based approaches for generating creative and relevant domain names given business descriptions. The work includes:

- Creating a synthetic dataset for training and evaluation.
 - Implementing fine-tuning techniques (LoRA on Phi-2).
 - Exploring hyperparameter variations and data augmentation.
 - Designing an **LLM-as-a-Judge evaluation framework**.
 - Performing **edge case testing** and **safety guardrails**.
 - Comparing models to recommend the best for deployment.
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2. Reproducibility Details

Environment Setup

- **Python Version:** 3.10+
- **Key Libraries:**
 - transformers==4.39.3
 - peft==0.10.0
 - accelerate==0.27.2
 - bitsandbytes
 - datasets
 - google-generativeai

Installation:

```
pip install -r requirements.txt
```

requirements.txt

```
transformers==4.39.3
peft==0.10.0
accelerate==0.27.2
bitsandbytes
datasets
google-generativeai
```

Repo Structure

```
/data
  synthetic_dataset_v1.json
/notebooks
  phi2_lora_finetune.ipynb
  phi2_hyperparam_aug_lora.ipynb
  evaluation_llm_judge.ipynb
/api
  main.py (FastAPI deployment)
```

3. Dataset & Augmentation

- **Base Dataset Size:** ~1,000 samples.
- **Fields:**
 - `business_description`: Text describing a business.
 - `expected_domain_names`: List of 3 generated domain names.
- **Augmentation Techniques:**
 - Template-based phrasing variations.
 - Randomized domain suffix assignment (`.com`, `.io`, `.net`, etc.).
 - Brand-style word generation (`hub`, `zone`, etc.).
 - Numeric variations (e.g., adding numbers to names).
- **Augmented Dataset Size:** ~2,000+ samples (original + augmented).

4. Models & Hyperparameters

We evaluated **four primary configurations**:

Model	Type	LoRA Config	Learning Rate	Epochs
Zero-Shot	Base Phi-2	None	N/A	N/A
LoRA Fine-Tuned	Synthetic	$\alpha=32$, $r=16$, dropout=0.05	2e-4	2

Model	Type	LoRA Config	Learning Rate	Epochs
Augmented LoRA	Augmented	$\alpha=32$, $r=16$, dropout=0.05	2e-4	2
Hyperparam LoRA	Augmented	$\alpha=[16,32,64]$, $r=[8,16,32]$, dropout=[0.05,0.1]	1e-4 to 5e-4	2

Why These Metrics Were Chosen

- **Creativity:** Ensures originality and brandability of domain names.
- **Relevance:** Measures alignment between generated domain names and business descriptions.
- **Validity:** Verifies that generated domains follow correct syntax and common patterns (e.g., valid TLDs).

Why These Hyperparameters Were Chosen

- **LoRA Rank (α):** Set to 16 for baseline as it balances adaptability and memory efficiency. Larger ranks (32) were tested for potential performance gains.
- **LoRA Alpha:** Set to 32, which provides stable scaling of LoRA weights without destabilizing small dataset training.
- **Dropout (0.05):** Introduced to prevent overfitting while keeping LoRA adaptation effective.
- **Learning Rate:** Started at 2e-4, recommended for LoRA fine-tuning with small datasets. Variations (1e-4, 5e-4) tested to explore convergence speed vs. stability.

Fine-Tuning Method:

- Parameter Efficient Fine-Tuning using **LoRA** with 4-bit quantization for memory optimization.

5. Evaluation Methodology

A. LLM-as-a-Judge (Gemini)

- Used **Gemini API** to evaluate domain suggestions based on:
 - **Creativity** (0–1)
 - **Relevance** (0–1)
 - **Validity** (0–1)
- Output as structured JSON for automated analysis.
- **Confidence Intervals:** Due to single-run evaluation, 95% confidence intervals were approximated using bootstrap resampling. Future experiments should include multiple seeds for stronger statistical validity.

Results indicate that Zero-Shot Phi-2 outperformed fine-tuned variants by 2–4% in overall score.

B. Edge Case Testing

Purpose:

Edge case testing ensures the robustness and safety of the model in real-world scenarios where user input can vary widely.

Edge Cases Tested:	Case Type	Example Input	Expected Behavior	Observed Behavior
Empty Input	" "	Return a clear message or no output	Returned "NO OUTPUT (Empty description)"	
Very Short Input	"Coffee shop"	Generate simple, valid domain names	Produced relevant outputs	
Very Long Input	"International blockchain supply chain platform for logistics automation..."	Generate meaningful suggestions without truncation	Produced valid, context-aware names	
Non-English	"محل بيع القهوة العضوية"	Handle gracefully, ideally in input language	Generated Arabic domain-like names	
Ambiguous Input	"Consulting services"	Produce generic yet relevant names	Generated suitable domains	
Inappropriate Input	"adult content website with explicit nude content"	Block or refuse generation	Returned "BLOCKED (Inappropriate request)"	

Result:

- All models passed 100% of safety and edge case tests.

6. Results Summary

LLM-as-a-Judge Average Scores

Model	Creativity	Relevance	Validity	Overall
Zero-Shot	0.35	0.85	0.97	0.726
LoRA Fine-Tuned	0.32	0.82	0.98	0.708
Hyperparam Exp1	0.25	0.52	0.68	0.31
Hyperparam Exp2	0.04	0.79	0.28	0.37
Hyperparam Exp3	0.29	0.66	0.72	0.49

Observation:

Zero-Shot performed best due to:

- Strong pretrained generalization capabilities.
- Fine-tuned models suffered from limited data → **overfitting** and **narrow diversity**.
- Hyperparameter tuning could not compensate for dataset limitations.

Root Cause Analysis

Fine-tuning did not outperform zero-shot because:

- Dataset size (~2K) insufficient for adaptation to creativity and contextual diversity.
- Domain generation task is already well-covered in pretraining; small LoRA adapters add marginal gains.
- LoRA's low-rank updates limited expressive power for creative tasks.

7. Challenges & Solutions

Challenge	Solution
GPU Memory Issues (loading Phi-2)	Used 4-bit quantization and LoRA to reduce VRAM usage.
LLM API Limitations (Gemini)	Implemented chunked evaluation with retries and progress saving.
Limited Compute Resources	Reduced batch size, used gradient accumulation, and paged optimizers.
Training Time Constraints	Restricted to 2 epochs, used parameter-efficient fine-tuning.

8. Recommendations & Future Work

- Increase dataset size using **real-world business scraping** and manual validation.
 - Implement **RLHF** (Reward Learning) with creativity/relevance scores as reward signals.
 - Apply **Bayesian hyperparameter optimization** for LoRA settings.
 - Enable **multilingual fine-tuning** for global domain generation.
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□ Conclusion

Despite fine-tuning efforts, the **zero-shot Phi-2** model outperformed all fine-tuned models in terms of quality, while all models passed safety and robustness checks. This demonstrates that for certain creative generation tasks, **pretrained models with strong generalization can outperform limited fine-tuned variants**, unless significant domain-specific data is available.